

REUSABLE PROOF THAT A COMPLIANCE VERIFICATION IS STILL VALID



## Ask for the fact. Not the file.

A living registry of compliance attestations on Midnight: reconfirm that a partner's check is still valid, without ever receiving the file behind it.

### ATTESTATION

# . . . .

Originator

[REDACTED]

Date of birth

[REDACTED]

Address

[REDACTED]

ID document

[REDACTED]

Screening result

[REDACTED]

Screened against

[REDACTED]

*Raw data not received, by design.*

**LIVE**

10:00 • THE MOMENT THE PRODUCT EXISTS TO CHANGE

“I know I shouldn't ask for this. But I have no other way to confirm the fact.”

Priya Chandran  
Compliance & AML Lead, mid-sized VASP  
Composite persona, not a real person or testimonial

A payment crosses the FATF Travel Rule threshold. The rule needs one fact: the originator was screened and is not on a sanctions list. To confirm that one line, she asks for the whole file.

| TIME  | WHAT HAPPENS                                               | FEEL |
|-------|------------------------------------------------------------|------|
| 09:00 | Large transfer triggers the Travel Rule                    | +1   |
| 09:15 | Ask for too much, or hold the customer                     | -1   |
| 10:00 | Requests the raw file, knowing it breaks data minimization | -3   |
| 12:00 | Waits on the counterparty                                  | -2   |
| 14:30 | A PDF arrives: full name, address, ID                      | -2   |
| 16:00 | Approves, now custodian of data that isn't hers            | -1   |
| 17:00 | Files it. Next time, the same again                        | -1   |

# The rule is spreading faster than anyone can enforce it, and small institutions pay the most.



of the 91 jurisdictions surveyed have legislated the Travel Rule (73% in 2025). Only 40% enforce it.

FATF, 2026

US\$72.9M

average annual AML/KYC compliance spend per institution.

Fenergo via fintech.global, 2026

8.7% vs 2.9%

compliance share of non-financial expense, small institutions vs large ones.

FDIC

~6,000

institutions reuse each other's KYC in the SWIFT KYC Registry. A closed club: outside it, you ask for the whole file.

SWIFT KYC Registry

# An attestation says a fact was true once. It doesn't say it's still true today.

## ATTESTATION

A promise made once.

- Signed at one moment by the issuer
- Can go stale: a sanction hits, a licence is pulled
- Today it travels with the raw file behind it

## VERIFICATION

Trust, confirmed right now.

- Still inside its validity window, by real block time
- Not revoked by the issuer
- From an issuer on the trust list
- This is what Attesta proves, without the file

# Register once. Reconfirm on demand. Revoke, and the next proof says so.

|                                                                                                                                                                          |                                                                                                                                                                               |                                                                                                                                                                            |                                                                                                                                  |                                                                                                                                                     |                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <div>01 · ISSUER</div> <div>registerAttestation</div> <div>Only a commitment to the record goes on-chain. The file never touches the contract.</div> <div>ON-CHAIN</div> | <div>02 · LEDGER</div> <div>attestations tree</div> <div>The commitment becomes a leaf. Everyone sees the <b>root</b>, not which leaf is whose.</div> <div>PUBLIC STATE</div> | <div>03 · ISSUER → VERIFIER</div> <div>proof packet</div> <div>Handed over once. Hashes, salt and two dates. Never the issuer's secrets.</div> <div>OFF-CHAIN · ONCE</div> | <div>04 · VERIFIER</div> <div>proveLive</div> <div>Proves membership, window, trusted issuer, not revoked.</div> <div>LIVE</div> | <div>05 · ISSUER</div> <div>revokeAttestation</div> <div>Adds a nullifier to revokedNullifiers. The root does not change.</div> <div>ON-CHAIN</div> | <div>06 · VERIFIER</div> <div>proveLive again</div> <div>A new proof, a new answer. No call to the issuer.</div> <div>REVOKED</div> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|

**Reuse is the point:** the verifier can run step 04 as often as she needs, and anyone she shares the packet with can too, without the issuer being online.

# The verifier gets a status. Never the file.

Proof packet

Send this text to the verifier out-of-band (paste it into the verifier panel "Import proof packet" box). It never contains this issuer's secret, and never the raw record, only the hashed record fields `proveLive` already discloses, plus the commitment.

```
{
  "attestaProofPacketVersion": 1,
  "rawDataHash":
"8dab77235a6fd8eaad375b75bcf9239045c50fb673420ecfe252dfd84
  "validFrom": "1789550621",
  "validUntil": "1792142621",
  "issuerId":
"34ceebd7334686be2012ce83e08611f4204aad832324556b484155516
  "nullifierHash":
"0c37c4df17d259e9c2b06d4ac676434614bflba4ed1a4d87c83aa9ae4
  "salt":
"29eda43d66da911163ef1c6cc3fc1081efd1f2698f30808061034ca52
  "commitment":
"0ac99eeb2b11fe97b14223e41a014c0535a31a34e164a8eef030fcd36
}
```

COPY TO CLIP

1 · PROOF PACKET, HANDED OVER ONCE

Verifier

Import a proof packet from the issuer, then request a live, on-chain proof that the attestation is still valid, right now, not when it was issued.

Connected wallet: `mn_shield-...hnyqgskk9q`

Import proof packet

Paste the JSON the issuer exported...

IMPORT PROOF PACKET

Imported attestation  
Imported #1, local ref 1a5b47...76f087

The local ref is a selector this panel generated for its own bookkeeping, never the underlying document or the counterparty's identity, which this panel never received.

REQUEST PROOF

LIVERE-REQUEST PROOF

Last on-chain proof: 2026-09-16T09:24:40.144Z. Status above tracks the public ledger live (e.g. a revocation in the issuer panel updates it automatically). Use "Re-request proof" any time for a fresh cryptographic proof instead of the tracked read.

|             |                      |
|-------------|----------------------|
| Issuer id   | 34ceebd7...6330753e  |
| Valid from  | 2026-09-16T09:23:41Z |
| Valid until | 2026-10-16T09:23:41Z |

2 · PROVELIVE RETURNS LIVE

Verifier

Import a proof packet from the issuer, then request a live, on-chain proof that the attestation is still valid, right now, not when it was issued.

Connected wallet: `mn_shield-...hnyqgskk9q`

Import proof packet

Paste the JSON the issuer exported...

IMPORT PROOF PACKET

Imported attestation  
Imported #1, local ref 1a5b47...76f087

The local ref is a selector this panel generated for its own bookkeeping, never the underlying document or the counterparty's identity, which this panel never received.

REQUEST PROOF

REVOKEDRE-REQUEST PROOF

Last on-chain proof: 2026-09-16T09:25:50.807Z. Status above tracks the public ledger live (e.g. a revocation in the issuer panel updates it automatically). Use "Re-request proof" any time for a fresh cryptographic proof instead of the tracked read.

|             |                      |
|-------------|----------------------|
| Issuer id   | 34ceebd7...6330753e  |
| Valid from  | 2026-09-16T09:23:41Z |
| Valid until | 2026-10-16T09:23:41Z |

3 · AFTER REVOKE: REVOKED

# One door between the two sides, and the compiler guards it.

| PRIVATE · WITNESSES, NEVER LEAVE THE DEVICE |  |                            |
|---------------------------------------------|--|----------------------------|
| raw screening file                          |  | <div></div>                |
| issuerSecret                                |  | issuer only                |
| revocationSecret                            |  | issuer only                |
| salt, rawDataHash                           |  | issuer + packet holder     |
| Merkle path                                 |  | witness, fetched per proof |

disclose()

| PUBLIC · MIDNIGHT LEDGER |  |                                        |
|--------------------------|--|----------------------------------------|
| commitment leaf          |  | registerAttestation                    |
| tree root                |  | same for every member                  |
| nullifierHash            |  | revoke, and each proveLive lookup      |
| issuerId                 |  | proveLive result, trust check          |
| validFrom, validUntil    |  | proveLive block-time check             |
| status                   |  | LIVE · EXPIRED · REVOKED · NOT_TRUSTED |

**What we are not claiming.** Each proveLive is an on-chain transaction: the verifier pays DUST per check, and the fields it discloses make repeated checks of the same attestation linkable to each other and to a later revocation. The raw file stays private; the check itself is not invisible.

```
export ledger attestations:
  HistoricMerkleTree<10, Bytes<32>>;
export ledger revokedNullifiers:
  Map<Bytes<32>, Boolean>;
export ledger trustedIssuers:
  Map<Bytes<32>, Boolean>;

export circuit registerAttestation(ref): Bytes<32>
export circuit revokeAttestation(ref): []
export circuit proveLive(ref):
  [LivenessStatus, Bytes<32>]
export circuit setTrustedIssuer(issuerId, trusted): []
```

PROVE LIVE · STATUS, IN EVALUATION ORDER

|   |                                  |             |
|---|----------------------------------|-------------|
| 1 | nullifier in revokedNullifiers   | REVOKED     |
| 2 | outside validFrom ... validUntil | EXPIRED     |
| 3 | issuer not in trustedIssuers     | NOT_TRUSTED |
| 4 | otherwise                        | LIVE        |

**Revoking never changes the root.** It inserts one nullifier that is not derivable from the public commitment.

**Time comes from the chain,** via `kernel.blockTimeLessThan`, not from the caller.

**Writing needs the issuer's secret.** A verifier with only the packet can prove, never register or revoke.

# Take Midnight away, and two things break.

## 1 · DISCLOSURE IS A COMPILE-TIME RULE

**A leak is a build failure, not a runtime bug.**

```
witness rawDataHash(ref: Bytes<32>): Bytes<32>;  
export circuit leakRawAttestationData(ref) {  
    return rawDataHash(ref); // no disclose()  
}
```

Compact rejects witness data reaching public state without `disclose()`. This fixture sits in our test suite, which fails if it ever compiles.

## 2 · MEMBERSHIP, NOT IDENTITY

**Revocation with live state, built in.**



`root` · HistoricMerkleTree + revokedNullifiers map

Zero-knowledge membership proofs over a native Merkle tree, next to private witnesses. Elsewhere this is rebuilt by hand, in the exact place where a bug means accepting a revoked attestation as valid.

14/14 contract tests passing on the simulator

- ✓ registerAttestation discloses only a commitment
- ✓ revoked attestation is rejected by proveLive
- ✓ revoking one does not affect another
- ✓ expired and not-yet-valid windows are rejected
- ✓ time tracks real block time, not the caller
- ✓ verifier with only the packet can prove liveness
- ✓ fabricated issuerId fails closed
- ✓ leak fixture fails to compile (CI, compact 0.31.1)

Also in the repo: bboard-ui (issuer and verifier panels) and bboard-cli, wired to all four circuits.

## MIDNIGHT PREPROD

### CONTRACT ADDRESS

4f2cd18fd2c09aef3960f5159d29981fa4470a6bb26b2c1e0ce36537e6362f97

### DEPLOY TX

121807350a31fdf8d196c76c8377a5e8b86cf1c71cc3d975f3ef57f03082e277

### Reference contract: deployed, no calls yet

As of 15 Sep 2026 the public indexer shows only the deploy. Judges can join it from the live app with a funded 1AM wallet.

### Full cycle recorded on a local devnet

Deploy, trust, register, prove LIVE, revoke, prove again: REVOKED. Every step is a real transaction; see the demo video and docs /demo-walkthrough.

Built with compactc 0.31.1. Tested wallet on preprod: 1AM.

# What Wave 1 does not solve.

## SIMULATED TRUST LIST

Trusted issuers are demo participants. Issuer governance is not solved in this Wave.

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### Trust list writes are unauthenticated

`setTrustedIssuer` can be called by any wallet. Acceptable for a demo, not for production.

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### Repeated checks are linkable

`proveLive` discloses the nullifier hash, issuer id and validity window on every call.

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## SIMULATED SANCTIONS LIST

The sanctions screening type is a labelled stand-in, not a live OFAC, EU or UN feed.

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### Private state lives in memory

Reloading the browser loses the issuer's secrets and imported packets.

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### Who audits the verifier?

Not answered in this release. It is the question Wave 2 is built around.

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# Attesta will add an audit layer next.

DELIVERED

## WAVE 1

**Register, prove, revoke** in one Compact contract

Proof packet reuse without the issuer

Issuer and verifier panels, CLI, preprod deploy

NEXT · PLANNED

## WAVE 2

**Audit layer:** private verification receipts, and one aggregate proof that a quarter of checks followed policy, without opening a case

Will authenticate trust list writes

Will connect real issuer trust and a sanctions feed

Will reduce linkability of repeated proofs

LATER · PLANNED

## WAVE 3

Multi-institution audit

A policy engine institutions can configure

Production gated on a security audit

# Start where the cost hurts most and the closed club never reaches.

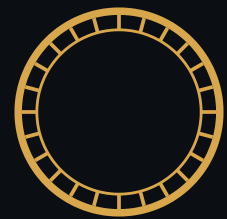
WHO GROUNDED

|             |                                                       |
|-------------|-------------------------------------------------------|
| Segment     | Small and mid-sized VASPs, exchanges, payments firms  |
| Outside     | the ~6,000-member SWIFT KYC Registry                  |
| Pressure    | Travel Rule: 83% legislated, 40% enforced (FATF 2026) |
| Cost weight | 8.7% vs 2.9% of non-financial expense (FDIC)          |
| Two roles   | Each institution is issuer and verifier               |

HOW THESIS

- |   |                                                         |
|---|---------------------------------------------------------|
| 1 | Pilot: one issuer, one verifier, sanctions screening    |
| 2 | API integration into existing compliance tooling        |
| 3 | A shared registry with a founding group of institutions |

Revenue options under consideration: per-verification fee, registry membership, self-hosted licensing. No pricing has been tested with an institution yet.



# Attesta

Ask for the fact. Not the file.

If you work in compliance, AML, or  
KYC, even informally, we want 15  
minutes of your time to try to break this.

[github.com/zzaved/Attesta](https://github.com/zzaved/Attesta)

Docs: [zzaved.github.io/Attesta](https://zzaved.github.io/Attesta) · App: [attesta-rx88.onrender.com](https://attesta-rx88.onrender.com)



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